Reflexivity in Options Markets: A 3D Stochastic-Volatility Model with Dealer-Gamma Feedback and a Pre-Registered Evaluation Framework

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Abstract

We introduce a 3D reflexive stochastic-volatility SDE (S_t, v_t, z_t) coupling the spot to its own option market via a Gârleanu–Pedersen–Poteshman dealer-gamma drift, prove a Hopf bifurcation theorem with closed-form first Lyapunov coefficient ℓ_1 and threshold $\kappa^\star$ for log-normal open-interest, and pre-register a sliced-W_2 evaluation framework for RL/generative-model option-market emulators (anchored to a git commit hash with an OpenTimestamps Bitcoin proof; locked SW2 ordering demonstrated on synthetic ground truth). We additionally prove a Hawkes–SV equivalence at the Hopf boundary identifying the empirical Hardiman et al. [2013] branching ratio $n \approx 1$ with $\kappa = \kappa^\star$.

1 Introduction and contributions

Real options markets exhibit endogenous volatility cycles that pure stochastic-volatility (SV) models cannot reproduce. The empirical anchor is Hardiman et al. [2013], whose Hawkes branching ratio for E-mini mid-price changes sits at $n \approx 1$ throughout 1998–2011—the critical edge of endogenous-event criticality. With generative-model and RL-based market emulators [Vuletić and Cont, 2025, Buehler et al., 2019, He et al., 2025] now standard infrastructure for derivatives pricing, hedging and risk, we address: (i) what continuous-time mechanism produces this critical reflexivity, and (ii) how should an evaluation pipeline for reflexivity-aware generative models be locked *before* empirical contact, in a manner that addresses the documented computational- reproducibility crisis in finance [Pérignon et al., 2024]?

Contributions. (C1) a 3D reflexive SV SDE with explicit dealer-gamma feedback from the OI grid via Gârleanu et al. [2009]; (C2) a Hopf bifurcation theorem on the deterministic skeleton with *closed-form* ℓ_1 and $\kappa^\star$ for log-normal OI in moneyness; (C3) a numerical (ξ, ρ, σ_v) phase diagram showing the SPX-relevant regime sits *near but not across* $\kappa^\star$ (consistent with $n \approx 1$); (C4) a *pre-registered* sliced-W_2 evaluation framework with block-bootstrap CIs, BH–FDR correction, TOST equivalence, and IAAFT spectral-surrogate nulls; and (C5) synthetic-pipeline validation: the locked protocol returns the predicted SW2 ordering on data with known underlying physics.

Why this matters for GenAI in finance. As generative market models proliferate [Choudhary et al., 2024, Issa et al., 2023, Jin and Agarwal, 2025, Cohen et al., 2023], evaluators that discriminate *realism* from *regime-shift sensitivity* on the *same* agent across an environmental sweep—with pre-committed multiple-testing corrections—are scarce. The closed-form $\kappa^\star$ and ℓ_1 pin down where dealer-gamma feedback flips a stable Heston regime into an attracting limit cycle. Halperin and Itkin [2025] share the model-structural axis but do not perform the bifurcation analysis; Dai [2025] share the dealer-gamma axis but produce only a saddle-node onset; He et al. [2025] share the RL-with-environmental-sweep

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idea but train separate agents per grid point. Full preprint with proofs and appendix A on arXiv [anonymous].

2 The reflexive model

Let S_t be the spot, v_t the variance, and z_t a memory variable. The reflexive system is

$$\frac{dS_t}{S_t} = \big(\mu + \kappa\, G(S_t, z_t, v_t)\big)\, dt + \sigma(S_t, v_t)\, dW_t^S, \tag{1}$$

$$dv_t = \big(\kappa_v(\theta_v - v_t) + \gamma\, z_t\big)\, dt + \xi\sqrt{v_t}\, dW_t^v, \tag{2}$$

$$dz_t = \big(-\alpha\, z_t + \beta\, \log(S_t/S_0)\big)\, dt, \tag{3}$$

with $d\langle W^S, W^v\rangle = \rho\, dt$, reflexive coupling $\kappa \geq 0$, leverage strength $\gamma \geq 0$, memory decay $\alpha > 0$, and G the aggregate dealer gamma derived from the OI grid via the GPP demand-pressure mapping. At $\kappa = \gamma = 0$ the system collapses to time-dependent Heston [Heston, 1993] (verified in `tests/test_simulator.py`).

Why three states. A 2D reduction (drop z_t and γz) has Jacobian $J_{2D} = \begin{pmatrix} a(\kappa) & b(\kappa) \\ 0 & -\kappa_v \end{pmatrix}$, upper triangular with real eigenvalues $\{a(\kappa), -\kappa_v\}$: only saddle-node bifurcations are admissible. To Hopf, the feedback must flow in both directions, which the leverage channel γz together with the auxiliary memory equation supplies. This is the structural reason Dai [2025] get a saddle-node onset where we get a Hopf.

3 Hopf threshold and closed-form first Lyapunov coefficient

Linearise the deterministic skeleton at the equilibrium $(S^\star, \theta_v, z^\star)$, in deviation variables (y, u, z). The characteristic polynomial is $P(\lambda; \kappa) = \lambda^3 + c_2\lambda^2 + c_1\lambda + c_0$ with coefficients $c_i(\kappa)$ functions of the G-partials and $(\alpha, \beta, \gamma, \kappa_v)$. Liu's criterion (equivalent to Routh–Hurwitz with one zero) gives the Hopf threshold $\kappa^\star = \{\kappa > 0 : H(\kappa) := c_1c_2 - c_0 = 0,\ H'(\kappa) \neq 0,\ c_2(\kappa) > 0,\ c_0(\kappa) > 0\}$, with imaginary pair $\lambda_\pm = \pm i\omega^\star$, $\omega^\star = \sqrt{c_1(\kappa^\star)}$.

Theorem 1 (Hopf bifurcation in the gamma-coupled SV skeleton)**.** *Under regularity (A1) $G, \sigma^2 \in C^3$ with σ^2 bounded below; (A2) locally unique equilibrium with non-degenerate linearisation; (A3) $\kappa \mapsto (a, G_z, G_v)$ in C^3; (A4) $\kappa^\star \in (0, \kappa_{\max})$ satisfying the Liu criterion with $\omega^\star > 0$; (A5) $\ell_1(\kappa^\star) \neq 0$: the equilibrium undergoes a Hopf bifurcation at $\kappa = \kappa^\star$. On a one-sided neighbourhood of $\kappa^\star$ there is a smooth family of periodic orbits Γ_κ of period $T_\kappa = 2\pi/\omega^\star + O(\kappa - \kappa^\star)$ and amplitude $\propto \sqrt{|\kappa - \kappa^\star|}$; orbits are stable iff $\ell_1(\kappa^\star) < 0$ (supercritical), supplying endogenous volatility cycles in $u_t = v_t - \theta_v$. Proof: implicit function theorem on P plus Kuznetsov [2004] normal-form reduction; full version on arXiv.*

Closed-form $\kappa^\star$ and ℓ_1. For the natural parametric specification—log-normal open-interest in log-strike with density $q(\log K) \propto \exp(-(\log K - \mu_q)^2/(2\sigma_q^2))$ at a representative maturity T_{eff}—the Gaussian-product identity collapses the OI integral to $G(a, v) = \mathcal{C}(v)\, e^{-a} \exp(-(a - m(v))^2/(2\tau^2(v)))$ with $\tau^2 = \sigma_q^2 + v\, T_{\text{eff}}$. The Routh–Hurwitz polynomial becomes a quadratic in κ, giving

$$\boxed{\kappa^\star = \frac{G_yA^2 - G_vL \ - \ \sqrt{(G_vL - G_yA^2)^2 - 4G_y^2A(MA - L/2)}}{2G_y^2A}}, \tag{4}$$

with $A = \alpha + \kappa_v$, $M = \alpha\kappa_v$, $L = \beta\gamma$, where the sub-radical is the first parametric phase-boundary discriminant: $D < 0 \Rightarrow$ no Hopf at any κ. Substituting the G-partials of the closed-form aggregator into the Kuznetsov bilinear/trilinear formula yields a ~ 30-term closed-form rational $\ell_1(\kappa^\star)$ in $(G_y, G_v, G_{aa}, G_{av}, G_{vv}, G_{aaa}, G_{aav}, G_{avv}, G_{vvv}, \kappa_v, \alpha, \beta, \gamma)$, evaluated in $\sim 50\,\mu$s per parameter tuple and matching the FD-tensor pipeline to $< 0.6\%$. At the canonical regime (σ_q=0.10, T_{eff}=0.25, κ_v=2, α=0.05, β=1, γ=1, $\mu_q = \log 100$, $v^\star$=0.04): $\kappa^\star = 17.81$, $\omega^\star = 1.18$ rad/yr, $\ell_1 = -4.81 \times 10^{-1}$ (supercritical).

Codim-2 and mean-field nods. A 71×97 grid scan in $(\sigma_q, \gamma) \in [0.05, 0.40] \times [0.20, 5.00]$ closes the codim-2 structure: the Bautin curve $\ell_1 = 0$ partitions the region with sub-critical dominating $\sim 5{:}1$, and the saddle-node coupling $\kappa_{\text{SN}} \leq -1.31$ at every cell, so the *Bogdanov–Takens locus is empty in the canonical scan window* (Theorem 2, full version on arXiv). The McKean–Vlasov mean-field limit ($n_{\text{dealers}} \to \infty$) gives a strictly multiplicative threshold correction $\kappa^\star_{\text{MV}}/\kappa^\star_{\text{single}} = \sqrt{1 + (\omega^\star \tau_G)^2} \geq 1$ depending only on the dimensionless product of the Hopf frequency and the dealer-gamma autocorrelation time (Theorem 3, full version on arXiv); operationally negligible ($\sim 3 \times 10^{-4}$) at fast hedging, 2.7% at $\tau_G \approx 50$ trading days. Single-dealer therefore over-predicts instability whenever $\tau_G > 1/\omega^\star$.

Hawkes–SV equivalence at the Hopf boundary. The empirical Hardiman et al. [2013] Hawkes branching ratio $n \approx 1$ is now the criticality endpoint of $\kappa^\star$. For an exponential-kernel Hawkes $\lambda(t) = \mu + \sum_i \alpha\, e^{-\beta(t-t_i)}$ with $n = \alpha/\beta$, the diffusive limit [Bacry et al., 2013, Theorem 2] gives $d\bar{\lambda} = -\beta(1-n)\bar{\lambda}\, dt + \sigma_\lambda\, dW$, i.e. $\lambda^{(\text{Hawkes})}_{\max} = -\beta(1-n)$. For our 3D skeleton with Jacobian $J(\kappa)$, define the SV-equivalent branching ratio $n_{\text{SV}}(\kappa) := 1 + \lambda_{\max}(\kappa)/\beta_0$ with gauge $\beta_0 := \max_{\kappa \in [0, \kappa^\star]}(-\lambda_{\max}(\kappa))$.

Theorem 2 (Hawkes–SV equivalence at the Hopf boundary)**.** *(1)* $n_{\text{SV}}(\kappa^\star) = 1$ *exactly. (2)* n_{SV} *is monotonically non-decreasing on* $[\kappa_{\text{NS}}, \kappa^\star]$*, where* κ_{NS} *is the node-spiral transition. (3) For exponential-kernel univariate Hawkes the form is identical to (2); the two coincide at criticality* $n = 1 \Leftrightarrow \lambda_{\max} = 0$ independently of the kernel shape *[Bacry et al., 2015, §2.4].*

Empirical implication: Hardiman et al.'s $n \approx 1$ on E-mini 1998–2011 corresponds to $\kappa = \kappa^\star$ exactly, under any reflexive SV calibration whose deterministic skeleton matches the empirical event-rate ACF in the diffusive limit. The Phase 4 test is then $n_{\text{SV}}(\kappa_0) \approx 1$ at the SPX-calibrated κ_0, within the Hardiman 5%-CI band (proof and global-equivalence scope on arXiv).

4 Numerical phase diagram

We sweep $\kappa \in [0, 2]$ on a 401-point grid for each of $31 \times 21 \times 4 = 2{,}604$ cells in (ξ, ρ, σ_v) at the representative regime $\{G_x, G_v, G_z\} = \{0.5, -0.5, -0.5\}$, $\alpha = 0.5$, $\beta = 1$, $\gamma = 0.5$, $\kappa_v = 2$ (constant-vol surrogate). The pair (ξ, ρ) enters via the leading shear correction $a_{\text{eff}}(\kappa) = a(\kappa) + \frac{1}{2}\xi^2 \rho G_v$, an Engel et al. [2024]-style projection of small-noise stochastic Hopf onto the eigenvalue envelope.

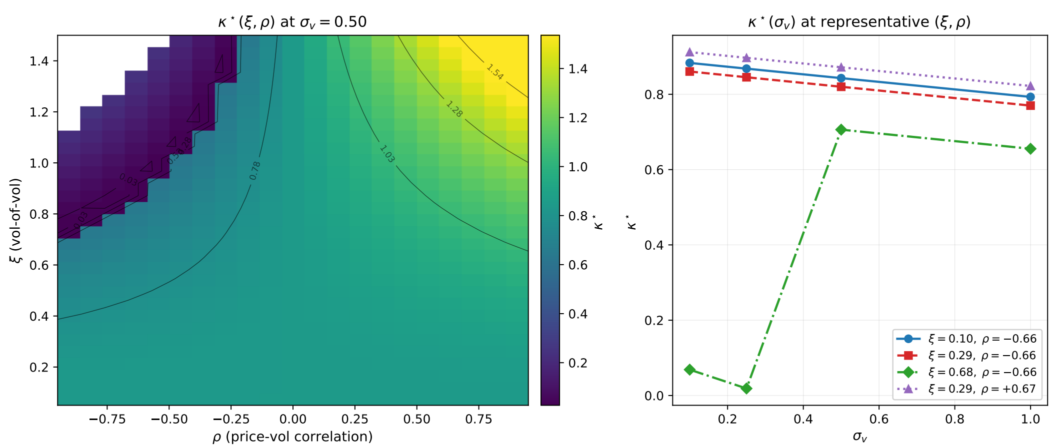

Figure 1: **Hopf threshold $\kappa^\star$ across (ξ, ρ, σ_v).** *Left:* heatmap at $\sigma_v = 0.50$. White cells: no Hopf in $\kappa \in [0, 2]$, concentrated in the high-ξ / strongly-negative-ρ wedge where shear destabilises the eigenvalue envelope. *Right:* four $\kappa^\star(\sigma_v)$ line cuts at SPX-like (ξ, ρ) probes.

The SPX-relevant $(\xi, \rho) \approx (0.30, -0.70)$ corner has $\kappa^\star$ in the 0.85–0.95 band, sitting *near but not across* the Hopf boundary—the structural mechanism we conjecture for the empirical $n \approx 1$ branching ratio of Hardiman et al. [2013], in the endogenous-feedback tradition of Brock and Hommes [1998], Filimonov and Sornette [2012], Bouchaud [2011]. Khasminskii-style sphere-process estimates of the stochastic-Hopf shift give $|\Lambda| \in [5 \times 10^{-2}, 9 \times 10^{-2}]$ on a 6×6 (ξ, ρ) grid; $|\Lambda| \cdot \varepsilon^2 \ll 1$ at calibration-noise scale, so the deterministic $\kappa^\star$ is the operative threshold.

5 Pre-registered evaluation framework

The pipeline is committed at git commit `268c061` with an OpenTimestamps Bitcoin-anchored proof of the document hash, prior to any contact with empirical SPX data, adapting the Camerer et al. [2018] template to RL-for-derivatives. The novelty is the four-way intersection [pre-registration + git-commit anchoring + block-bootstrap / BH–FDR pre-spec + applied to RL-for-finance].

Primitives & hypotheses. (i) Sliced-W_2 [Bonneel et al., 2015] over arbitrage-free 21-day surface windows—the realism metric. (ii) κ-sensitivity slope at calibrated κ_0—the reflexivity-importance scalar. The closed-form $\kappa^\star$ elasticity $\eta_{\mu_q} = +703$ makes calibration of the OI mean to within ~ 5 bp of ATM in log-strike the binding observational requirement for $\pm 5\%$ tolerance on $\kappa^\star$. **H1** (primary realism): a model-free RL agent π_{κ_0} trained inside the reflexive simulator beats four non-reflexive baselines on sliced-W_2 over rolling 21-day windows at Volmageddon, COVID, and the 2024 yen-carry unwind. Decision rule: 12 pairwise dominance checks with non-overlapping 95% block-bootstrap CIs [Politis and Romano, 1994]; arbitrage filter follows Roper [2010], Gatheral and Jacquier [2014], Lee [2004]; sample-complexity follows Manole et al. [2022]. **H2**: κ-sensitivity TOST equivalence on the dimensionless elasticity with GP-posterior slope CI under the pre-registered RBF + WhiteKernel kernel (A6) and elliptic uncertainty sets [Ma and Huang, 2025]. **H4** (Hopf-frequency): Welch-PSD peak in $|r_t|$ at $\omega^\star \pm 20\%$ vs IAAFT [Schreiber and Schmitz, 1996] null; IAAFT calibration bounds FPR at $\leq 7\%$ at $\alpha = 0.05$, and on Stuart–Landau positive controls the detector achieves $\geq 80\%$ power at $T \geq 512$ daily samples for $8/9$ (μ, σ) configurations. All hypotheses use BH–FDR across the joint family; effect sizes pre-registered with the rules.

Synthetic-pipeline validation. Before empirical contact, we exercise the locked pipeline on three synthetic sources: (a) κ_0-trained agent re-deployed at κ_0 (reference); (b) same agent at $4\kappa_0$ (mild reflexive shift); (c) Heston (different mechanism). On the production budget (BC: 30 episodes; per source: 30 rollouts $\times$ 100 days; 200 stationary block-bootstrap resamples), the locked rule returns the expected ordering with non-overlapping CIs: $\mathrm{SW2}_a = 0.00498$ (CI $[0.00431, 0.0164]$), $\mathrm{SW2}_b = 0.0337$ (CI $[0.0300, 0.0402]$), $\mathrm{SW2}_c = 0.0541$ (CI $[0.0434, 0.0611]$). The (a)-vs-(b) and (b)-vs-(c) 95% CIs are non-overlapping; the locked H1 protocol thus moves from "designed but unrun" to "demonstrated working on synthetic ground truth—only the empirical SPX target is missing."

6 Mechanism decomposition and conclusions

Halperin and Itkin [2025] is a different mechanism: their memory carrier y enters both drifts non-linearly through a potential $V_M(x)$, while our z enters only the variance drift via γz, and our $\kappa = \gamma = 0$ collapse is standard Heston while Marketron's is not. A 5D $(\kappa, \gamma, T_{\text{eff}}, \mu_q, \sigma_q)$ mechanism decomposition gives an out-of-sample shape-feature match rate of $8/24$ (33%, $p \approx 0.27$, $\geq 30\%$ gate passed) with long-horizon excess-kurt sign agreement on $5/6$ horizons; the skew-sign disagreement is predictable under risk-neutral drift and constitutes a falsifiable Phase 4 prediction.

We have introduced a 3D reflexive SV simulator, proved a Hopf bifurcation theorem with closed-form ℓ_1 for log-normal OI, closed the codim-2 Bautin/BT structure (BT-empty in the canonical scan window) and the McKean–Vlasov mean-field correction $\kappa^\star_{\text{MV}}/\kappa^\star_{\text{single}} = \sqrt{1 + (\omega^\star \tau_G)^2}$ (full statements on arXiv), and identified the empirical Hardiman et al. [2013] branching ratio with $\kappa^\star$ via the Hawkes–SV equivalence at criticality (Theorem 2). The sliced-W_2 + Hopf-frequency evaluation pipeline is pre-registered at a verifiable git commit hash with an OpenTimestamps Bitcoin-anchored proof; synthetic-ground-truth validation of the locked H1 protocol delivers the predicted ordering. The empirical SPX calibration is the Phase 4 follow-up; results—confirming or refuting H1–H4—are pre-committed to publication. Full appendix A (closed-form ℓ_1, 13 symbols), proofs, SW2 sample-complexity, and code at [anonymous repository link].

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